

McKENNA PARKER

Philadelphia, PA | 570-600-3410 | mckennarparker@gmail.com | [LinkedIn](#) | [Website](#)

EDUCATION

University of Pennsylvania – GPA: 3.69/4.0

Philadelphia, PA

B.S. Computer Graphics (Digital Media Design); Minor in Gender, Sexuality, & Women's Studies

May 2027

- **Relevant Coursework:** Interactive Computer Graphics; Procedural Computer Graphics (Current); 3D Modeling; Computer Animation (Current); Linear Algebra; Data Structures & Algorithms; Computer Systems; Computation Theory (Current)

SKILLS

Programming/Markup Languages: C++/C#/C; Java; JavaScript; HTML/CSS; Python; SQL; OCaml

Frameworks/Libraries: OpenGL; GLSL; Java Swing; JUnit 5; Tailwind; Streamlit; Pinecone; LangChain; PyGame; NumPy

Tools: Qt; Visual Studio/VSCode; Unity; Git/GitHub; Blender; Maya; SolidWorks; Jira; IntelliJ

RELEVANT EXPERIENCE

Virtual Reality Software Engineering Intern

Philadelphia, PA

Penn Medicine/University of Pennsylvania CURF

June 2025 – Present

- Collaborated with a team of 5+ developers and surgeons to develop a **VR training didactic** for emergency airway surgery using the **Meta Quest 2 and 3**, **Unity Game Engine**, **C#**, and **Blender**
- Researched and developed a **RAG-enabled AI web app** that acts as a reference tool for airway trauma surgery using **Python's Streamlit**, **OpenAI**, **LangChain**, and **Pinecone** libraries

Academy of Math and Programming Scholar

New York, NY

Jane Street Capital

July 2023 – August 2023

- Designed solutions to **graph theory**, **game theory**, and **linguistics** problems in **Python** improving algorithmic thinking skills
- Collaborated with International Math Olympians during mathematics coursework in **combinatorics** and **number theory**
- Achieved 9th highest profit of 40 teams in the Electronic Trading Challenge by designing efficient ETF trading algorithms

MIT Women's Technology Program Attendee

Remote

MIT EECS Department

June 2022 – July 2022

- Selected as 1/20 from 300+ applicants to learn electrical engineering and computer science from current MIT students
- Developed an SIR disease model simulation game using **Python's PyGame**, **NumPy**, and random libraries
- Built complex circuitry projects, including an infrared heartbeat monitor and an aluminum foil piano powered by Arduino

PROJECTS

Mini Minecraft | C++, GLSL, OpenGL, Qt, Git/GitHub, OBS Studio

April 2025

- Collaborated with two peers to build a custom voxel game engine implementing **procedural terrain generation**
- Implemented efficient **visual hull rendering** and **chunking** and **multithreaded terrain generation** with GPU

3D Mesh Editor | C++, GLSL, OpenGL, Qt, Git/GitHub

March 2024

- Implemented OBJ file loading and editing using the **half-edge data structure** as seen in industry software such as Maya
- Developed functionality for **vertex deformation**, **face triangulation**, **edge-splitting**, and **Catmull-Clark subdivision**

2D and 3D Rasterizer with Camera | C++, Qt, Git/GitHub

February 2024

- Wrote a program that uses **triangulation**, **barycentric interpolation**, and **z-buffering** to draw a 2D image from a JSON file
- Expanded to 3D with **perspective-correct interpolation** of Z coordinates and **UV coordinates for texture mapping**
- Created a 3D perspective camera class using **view and projection matrices** allowing users to move and rotate the scene

LEADERSHIP EXPERIENCE

Venus Club Ultimate Frisbee

Philadelphia, PA

Social Captain

June 2025 – Present

- Organize weekly social events and outings for 40+ team members to encourage team bonding and relationship building

Division 1 Team Member

August 2023 – Present

- Earned one of twenty bids from 134 teams to the USA Ultimate Division 1 College Championship tournament in 2025
- Balanced three two-hour practices a week and 3+ weekend tournaments each semester with a full academic course load

Camp Kesem

Philadelphia, PA

Gold Unit Lead Counselor

August 2025

- Led four counselors and seven campers day-to-day, managing group and individual needs for an enjoyable, safe experience

Accessibility & Accountability Co-Director

August 2024 – May 2025

- Recruited Associate Director of campus LGBT Center to discuss gender inclusivity at camp with 80+ members
- Organized and presented a counselor training on neurodiversity and mental health to 50+ counselors

CERTIFICATIONS & AWARDS

Certifications/Courses: Intermediate Web Development (CodePath), Unity Essentials (Unity Learn), JavaScript Algorithms and Data Structures (freeCodeCamp), Responsive Web Design (freeCodeCamp), React (Codédex)

Awards: Penn Undergraduate Research Mentorship Awardee, Liberty Mutual Women in Tech Summit Attendee, National Merit Scholar Finalist, Selinsgrove Area High School Salutatorian